Feasibility Analysis & MVP Roadmap

Project: Climate Response Hub

Team: Program Counter

1. Executive Summary

Currently, emergency coordinators at Toronto Emergency Management (TEM) and Toronto Shelter & Support Services (TSSS) face significant cognitive load during extreme heat events, as they must mentally connect fragmented data across multiple tabs, such as the Heat Vulnerability Index and the Cool Spaces Near You map.

Climate Response Hub is a decision-support dashboard that integrates climate risk data and shelter access gaps into a single, unified view. By enabling coordinators to log their dispatch actions and track real-time status, the platform serves as a collaborative tool to prevent duplicate deployments and maximize the efficiency of relief resources.

2. Feasibility Analysis

This solution meets the practical business and technical feasibility criteria required by the hackathon, directly addressing the goals of municipal and utility partners.

A. Operational Feasibility

- Preventing Duplicate Efforts: When a coordinator logs a specific action on the dashboard (e.g., requesting 20 mobile cooling units for Mitchell Field Community Center in Willowdale East), the neighborhood is immediately marked, and other coordinators can update the status to 'Deployed' once the units arrive. This transparent tracking eliminates confusion, double deployments, and wasted resources among different departments.
- Reduced Decision-Making Time: By eliminating the "tab-switching" cognitive load, the platform accelerates crisis triage, transforming hours of manual data cross-referencing into seconds of actionable insights.

B. Economic Feasibility

- Resource Optimization: Efficiently targets limited physical relief budgets and resources to high-risk areas with the highest vulnerability scores (e.g., a score of 81) and severe accessibility gaps.

- Cost-Effective Architecture: The core infrastructure costs are extremely low. It leverages existing public data ecosystems (e.g., the City of Toronto's Heat Relief Network) and Esri's infrastructure, requiring very low initial capital for deployment.

C. Technical Feasibility

- Lightweight & Scalable MVP: For rapid prototyping and reliable performance, the current MVP uses React's state management and localStorage to simulate a real-time collaboration database without the overhead of maintaining heavy backend servers.
- Advanced Spatial & Weather Computing: Uses the Open-Meteo API to provide critical 12-hour forecasts, while the ArcGIS Spatial Analysis API actively calculates precise distances to cooling centers and identifies localized high-risk hotspots.
- Flexible Data Strategy (Real + Mock): While climate risks use real data, the platform handles real-time infrastructure status by using Mock APIs for cooling center capacity (e.g., 70% full) and backup power visibility. This design allows easy swapping of mock endpoints with live APIs from utility partners in the future.

3. MVP Scope & 6-Month Roadmap

Instead of overbuilding, our roadmap focuses on a minimum viable product that immediately solves the core collaboration problem, followed by strategic expansions.

- Phase 1: Current MVP (Months 0-1)
  - Integration of Esri GIS Heat Vulnerability Index mapping and Open-Meteo 12-hour forecasts.
  - ArcGIS Spatial Analysis to calculate exact distances to the nearest cooling centers.
  - Implementation of the collaborative CRUD workflow using React localStorage (e.g., Track Status: Pending ➔ Deployed).
- Phase 2: Data & Partner Integration (Months 2-3)
  - Transition from localStorage to a robust database (e.g., MongoDB) with strict Role-Based Access Control (RBAC) to securely handle sensitive emergency operations data. No individual resident data will be stored, ensuring strict privacy.

    - Transition from mock infrastructure data to actual utility APIs (e.g., Alectra Utilities) to monitor live power grid and backup generator statuses during extreme weather events.
- Phase 3: Service Expansion (Months 4-6)
    - Expansion beyond heatwaves to address winter cold snaps and early warning modules for other climate risks such as severe snowstorms and flash floods.
    - Development of automated Energy Equity reporting modules to assist utility partners with long-term infrastructure upgrade planning.

4. Strategic Impact

The Climate Response Hub goes beyond a simple monitoring tool; it serves as a critical infrastructure piece to achieve national and corporate resilience targets.

- Alignment with Government of Canada Goals: Directly supports the national target ensuring that by 2026, 80% of health regions will have implemented evidence-based adaptation measures, and strictly aligns with the ultimate goal to eliminate extreme heatwave deaths by 2040.
- Supporting Alectra Utilities: By providing unprecedented visibility into backup power and community vulnerability, the platform actively supports Alectra's climate resilience and emergency preparedness goals.